



Captain Marvel Post Credits

Marvel Studios held a press screening for Captain Marvel where they revealed the movie will have two post credit scenes. <https://digit.in/mar19-58>



AR mode on 'Pokemon Go'

The Pokemon Go update allows players to take photos of their Pokemon in the real world. <https://digit.in/mar19-59>

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he technological progress that humans have seen over the past two decades have turned a number of science fiction concepts into reality. These technologies include armoured and powered exoskeletons, self driving cars, watches with communication capabilities and AI assistants. All of us truly live in magical times. In this article, we have a look at some of the most common technologies in science fiction stories, and find out how close scientists are to actually realising them, and if there are any real scientific breakthroughs on the horizon.

ENERGY WEAPONS

Energy weapons are science fiction

staple. The ray guns in Star Trek and the blasters in Star Wars are examples of energy weapons in fiction. Many countries around the world are developing direct energy weapons, which use lasers, microwaves or accelerated particles. Dynetics and Lockheed Martin are developing the High Energy Laser Tactical Vehicle Demonstrator (HEL TVD), a 100 kW laser weapon mounted on a truck. The HEL TVD is meant to fire at incoming missiles, mortar rounds, artillery shells and hostile drones. The primary advantage of the laser weapon is the low cost of firing every round. In 1984, Russia developed what is known as the Soviet laser pistol, which was a hand held direct energy weapon for use in space by cosmonauts. The pistol was

designed to disrupt the optical instruments of enemy spacecraft. In India, the Defence Research Development Organization has collaborated with Bhabha Atomic Research Centre to develop an energy weapon known as Kilo Ampere Linear Injector (KALI). KALI is a cannon that fires pulses of accelerated electrons, and is designed to pulverise UAVs, satellites and any missiles that enter Indian airspace. Energy weapons are all still in the experimental stage, and at best are meant to complement existing kinetic weapon systems.

SPACE ELEVATOR

In 1895, a Russian scientist by the name of Konstantin Tsiolkovsky visited Paris, and was impressed by the Eiffel Tower. He imagined such a tower continuing all the way into space. In his mind's eye, he envisioned that at the end of the spire, there could be a celestial castle in space. Arthur C Clarke explored the idea in detail in his 1978 book, *The Fountains of Paradise*. In a fictional kingdom set in the Indian Subcontinent, a space elevator or orbital tower is constructed. Such a structure would provide cheap and easy access to space. The height of the space elevator would extend beyond geosynchronous orbit, and have a counter-weight which would be constantly pushed away from the Earth because of centrifugal force. A cable tether-

SCIENCE FICTION TO SCIENCE FACT

The miracles of modern science

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The HEL-TV concept by Lockheed and Dynetics

